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Colour measurement of foods by colour reflectance

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4.1 Introduction: food colour and quality

The first judgement of a food's quality is more often than not dependent on its various appearance characteristics, such as colour, surface structure and shape. Colour, in particular, is an important sensory attribute. It may be defined as the sensation experienced when energy, in the form of radiation within the visible spectrum, falls upon the retina of the eye (Francis and Clydesdale 1975). But this definition does not give the full picture. It is known that the brain further processes and conditions the signals received from both the rod-cells and the three different types of cone-cells on the retina, in order to rationalise the colour of an object despite changes in the level and colour of the surrounding lighting. This process of adaptation allows objects to be recognised as having virtually the same colour under very different conditions and is known as colour constancy. Over the last few years, many workers have tried to understand and develop models on constancy but as yet it remains an approximate quality. This is one of the reasons for resorting to instrumental methods for measuring colour under standardised conditions.

The food industry uses colour measurement for a number of reasons, all aimed at reducing the variability caused by subjective analysis and providing numerical specification. Firstly, there is a need to ensure a good colour quality in the product going to the consumer. Pangborn (1967) stated that 'In foods, colours are identified with previously experienced quality and serve as instant indicators of good or bad, according to the product and its intended use.' Consumers associate an acceptable colour with acceptable flavour, safety, nutrition and level of satisfaction (Christensen 1983). A perceived inferior colour quality or a colour variation from unit to unit of a branded product are factors not well appreciated by the consumer. The manufacturer, therefore, has the difficult task of providing a product of the expected colour quality at perhaps three different times of judgement: at purchase point if the product is packed in a transparent container; at the unpacking point in the kitchen; and at the consumer's plate after preparation and serving.

The first judgement, in the retail outlet, is particularly important because there are often many reference colours available in the form of competitor products, labelling

decor, advertising graphics, etc. In the case of products unwrapped in the home, colour expectation and colour memory become significant factors in the quality judgement. The colour of the food material on the consumer's plate is influenced by the manner of in-home preparation, in particular by the cooking procedure.

A second use for colour measurement is in the development of new foods or improved versions of existing foods, in cases where the colour is one of the major quality attributes. Here the reference would be the original product, and the aim would be to improve colour either by changing the manufacturing process, or by incorporating better colour-imparting ingredients in the recipe, or by reducing colour deterioration during storage. Colour measurement may also be used for the estimation of pigment concentration or tinctorial strength. Clydesdale (1977) has pointed out that in certain situations it is not possible to use optical density readings at absorbance maxima to do this job and that colorimetry often gives a better result.

In cases where colour is an important indicator of quality, colour measurement can provide a buying criterion in the purchase of certain raw materials or semifabricates. Examples of this application are in the purchase of tomato paste or citrus juices, where the colour gives an indication both of the original fruit quality and of the efficiency of the manufacturing process. Recent advances in instrumentation have increased the use of colour measurement for quality control and process optimisation. Simple 'pass/fail' judgements may be made both during manufacture and on the end product itself. Setting tolerance limits, in combination with sensory studies, allows greater control and, if well managed, can facilitate a greater uniformity of production between batches and even different production sites. Finally, it is possible to monitor a process by careful interpretation of colour measurement results. For example, a batch of ketchup measured more yellow than its target may suggest that it has been cooked for too long a time (Mabon 1993).

There are a bewildering variety of methods and instruments available to the food technologist in the field of colour measurement. When one is approaching the subject for the first time or when attempting to devise a method for a material outside the normal experience, the wealth of possibilities available sometimes make the choice difficult. It is the purpose of this chapter to attempt to identify a systematic approach in order to ease the task. The approach is concerned primarily with the use of tristimulus colorimetry and reflectance spectrophotometry.

4.2 Colour measurement principles: brief introduction

For a detailed treatment of the principles of colour measurement, the reader is referred to Chapter 3 in this book by D. B. MacDougall. Here we give a brief, practically oriented introduction to the subject.

4.2.1 Tristimulus colorimetry

The scientific basis for the measurement of colour is the existence of three different types of response signals in the human eye. Though four different types of receptor – ρ, γ and β cones, and rods – have been identified, the messages from these are encoded (in a way still not fully understood) to give three types of signal.

In historic work carried out in the late 1920s and early 1930s by Wright (1969) and Guild (1931), the performance response characteristics of a standard human eye (the

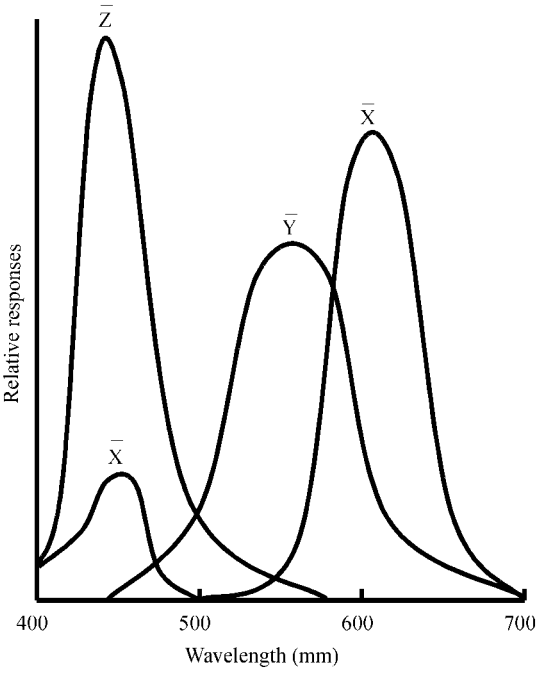


Fig. 4.1 Colour matching response functions of the human eye for the original CIE 2º observer.

standard observer) to different spectrum colour light sources were established. This work formed the basis of the Commission Internationale d'Éclairage system (CIE 1931). In order to be able to relate the results obtained from different combinations of primary colour stimuli, linear transformation equations were used to convert Wright and Guild's curves to more usable functions. This resulted in the definition of three integral primaries, *X*, *Y* and *Z* (see Fig. 4.1). These primaries do not exist as real lights, but they encompass all colours and are, therefore, mathematically very useful.

In a tristimulus colorimeter, three or four filters duplicate the response of the standard observer. The filters correspond to the three primary colours in the spectrum (red, green and blue) and they can be combined to match most colours. The more sophisticated instruments carry a fourth filter to simulate the blue part of the CIE [[*X* bar]] function shown in Fig. 4.1. The other essential parts of a tristimulus colorimeter are a white light source, an array of photometers, and an integral calculator (see Fig. 4.2a). Nowadays, the integral calculator is usually a computer which can also carry out data transformations between CIE and other colour scale systems or between different standard white light sources or white diffusers. Hunter and Harold (1987) give a good summary of the transformation formulae employed.

4.2.2 Reflectance spectrophotometry

Unlike the colorimeter's filter system, a spectrophotometer works by measuring the ratio of reflected light from a sample to that from a known reference standard. Measurements are taken, via an integrating sphere and then a diffraction grating, at many points across the visible range of the electromagnetic spectrum, i.e. between 380nm and 700nm (see Fig 4.2b). Different instruments measure at different wavelength intervals, depending

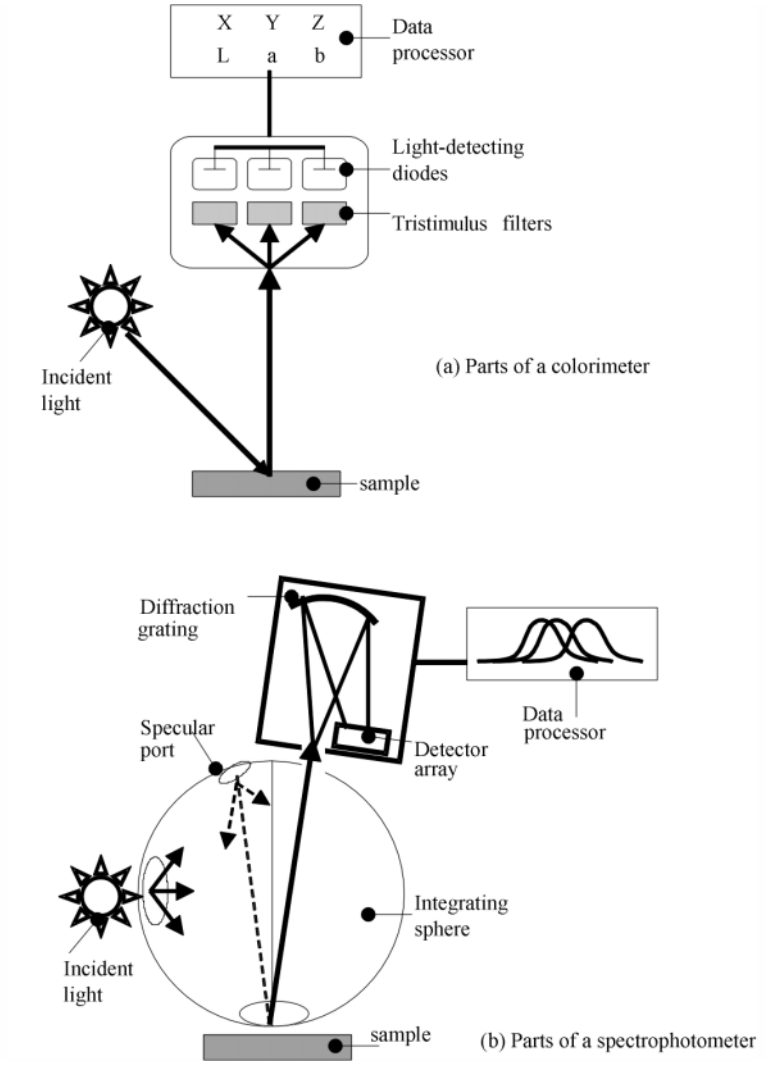


Fig. 4.2 Instruments for colour measurement by reflectance.

upon the nature of the spectral analyser and control electronics; 10nm or 20nm are accepted standards. Since reflectance is calculated by means of the above ratio, it is commonly expressed as a percentage. Thus, a perfectly reflecting diffuser will have a reflectance of 100%. During measurement analysis, however, reflectance is usually expressed as a fraction, with the ideal white tile then having a value of 1. Conversely, a black sample, which absorbs all incident light, will have a reflectance of 0% or 0. In between these two extremes fall the other visible colours. If a sample absorbs all but red light, it will show high reflectance values in the 'red' region of the reflectance spectrum. A spectrophotometer 'captures' all surface-reflected light across the visible wavelength range and thus can graphically analyse and output what the perceived colour should be. Every surface colour can, thus, be characterised by a reflectance curve.

An important characteristic of modern spectrophotometers is their dual-beam nature. The use of the second, or reference beam, is important in ensuring stability and

minimising error. The reference beam is light from the sphere walls which is collected at a reference port and directed via a mirror to another spectral analyser identical to the first. The resulting reflectance value output from the spectrophotometer, therefore, is a ratio rather than an absolute value and any drift resulting from electronic or physical error will be cancelled out.

Measurement error is often encountered during the definition of the measurement geometry, that is to say, the relationship between the incident (source) and reflected (measured) light. Most spectrophotometers are sphere-based instruments that allow the reflected light to be measured either with or without its specular component. As it is the surface properties of the sample which determine the amount of specular reflection, different results will be obtained when measuring a very rough surface as opposed to a highly gloss one. There is particular need for caution if comparing samples measured through a transparent packaging material. Excluding the specular component may result in the nature of the packaging determining the measured colour value. Including the specular light (i.e. keeping it within the integrating sphere) will ensure that only the colour and no surface effects are measured.

Despite the greater cost of a spectrophotometer in comparison to the simpler colorimeter, the ability to measure full reflectance curves allows colour-matchings to be made, tolerance volumes to be defined and a more in-depth understanding of many colour change processes to be achieved. Several manufacturers produce their instruments in both bench-top and portable models facilitating both lab analysis and near-line studies.

4.2.3 In-line colour measurement

The two commonly used instruments discussed above allow both laboratory and near process line colour measurement. With the advent of almost fully automated lines, true in-line colour measurement would be of great value in many factories. McFarlane (1988) discusses the economic benefits of in-line and closed-loop systems. His paper highlights closer adherence to specifications, minimisation of waste during start-up and product change, as well as an increase in process efficiency and plant throughput as key issues.

Work at the laboratory of one of the authors (PJ) has recently focused upon the use of fibre-optic reflectance probes to generate this in-line data. The fibre-optic cables may be bundled so as to provide both an illumination source and collection point in one probe (see Fig. 4.3). This probe is connected to a defined light source, such as a Xenon pulse providing D65, and then output to a spectrophotometer to generate full reflectance curves. The inclusion of a 45º optical window facilitates the 0º/45º viewing geometry. The whole probe is then inserted into a pipe to measure the product during its actual processing. No sampling procedure is required and with the appropriate software, instantaneous CIELAB or XYZ data may be acquired. As with all colour measurement systems, calibration is of key importance, as is the need to ensure that the pipe-fitting provides an adequate optical path-length without the interference of erroneous light. Such in-line systems have been previously used in the petroleum industries but with further investigation and development their routine integration into food processing could be a valuable future tool for both quality control and process validation.

4.2.4 Colour scales and colour difference formulae

The now routine integration of a pc means that the data collected from either a colorimeter or spectrophotometer can be transformed to XYZ, CIELAB, HunterLab or

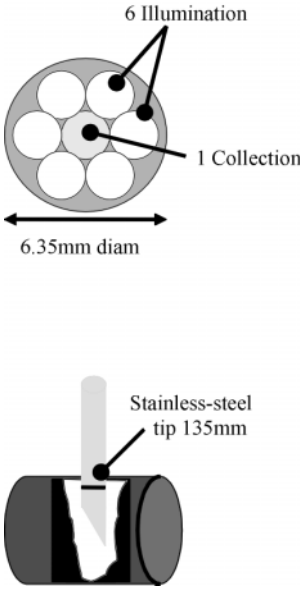


Fig. 4.3 A fibre-optic bundle composed of seven fibre-optics encased to make a probe for in-line colour analyses.

other colour system as specified by the user and provided by the software. The initial step from instrument response to XYZ requires the definition of standard observer functions.

The first standard observer was defined by the CIE in 1931. This was based upon a series of colour matching experiments carried out four years earlier and involved using a 2º field of view. This was thought to be adequate since in the human eye, the receptor cones, which are responsible for colour vision, were concentrated at 2º around the fovea. It was later established that this was, in fact, quite unrealistic as most visual assessments are done with a field of view much greater than 2º. Thus, in 1964 the CIE introduced the 10º Standard Observer – the 10º encompasses the area in the fovea that contains the rod receptors (responsible for lightness discrimination) and so is closer to human vision.

Many earlier studies have been carried out using the 2º observer and so caution must be exercised if comparing these values to those measured using the now more common 10º standard observer. Most software packages have the capabilities to convert data between the two; for those programming their own systems the raw data (chromaticity co-ordinates and colour matching functions) are available in the international standard ISO/CIE 10527 'CIE standard colorimetric observers'. The aim of the more popular of the colour scales is to represent colours in a three-dimensional colour space, such that similar visual colour differences are represented by approximately similar distances in the colour space. Such colour scales are known as uniform scales. The original CIE scales, represented diagrammatically in Fig. 4.4, were not intended for identifying the colours of objects and are certainly not uniform in spacing colours according to their visual differences. One of the more popular of the uniform colour scales in use in the food industry is the Judd-Hunter *L*, *a*, *b* solid, depicted in Fig. 4.5.

Hunter (1958) elaborated this scale with three purposes in mind; to be able to sense the scale directly with a tristimulus colorimeter; to provide terms which represented the red/green, blue/yellow and dark/light responses of the human brain; and to give a colour solid

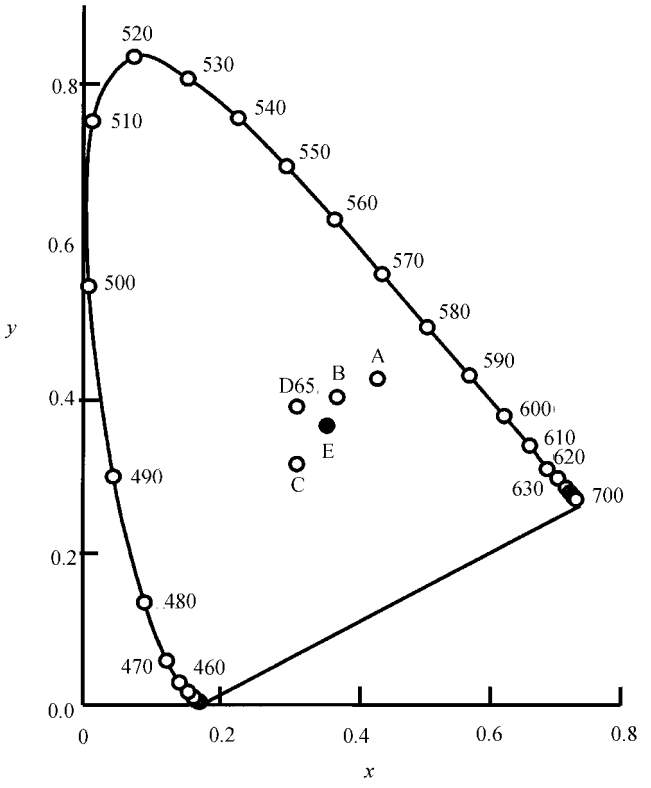


Fig. 4.4 CIE *x*, *y*, chromaticity diagram: locations of illuminants A, B, C, D65 and E (equal energy) are also shown.

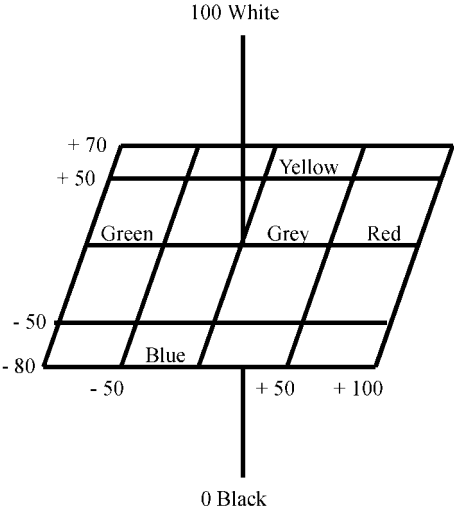


Fig. 4.5 Judd-Hunter colour solid.

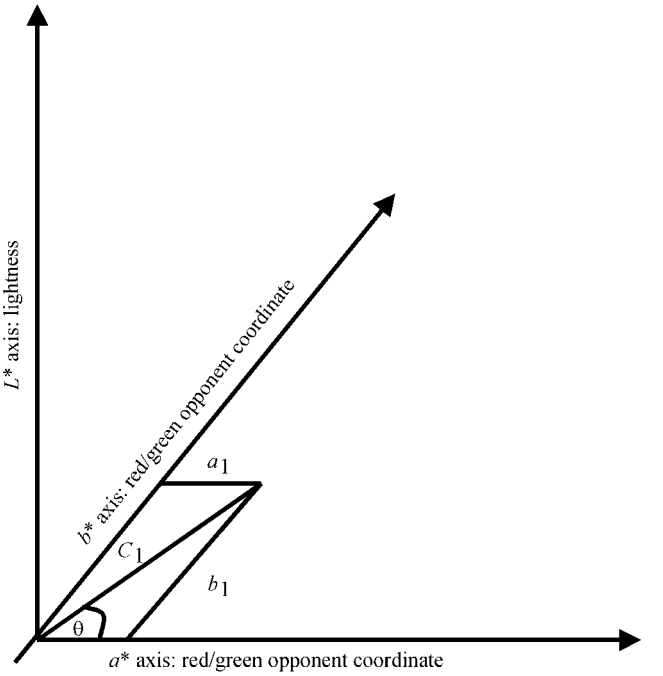


Fig. 4.6 Representation of the hue angle θ formed in the a^*-b^* plane by the a^* axis and the line from the origin to a point (a_1, b_1).

with visual uniformity across the solid. The latter aim is probably impossible to achieve, but the *L*, *a*, *b* scale comes close, as has been demonstrated by Hunter and Harold (1987).

In the Judd-Hunter solid, a colour can be represented by *L* (lightness or darkness, 0 = perfect black to 100 = perfect white), *a* ($-a$ = greenness, $+a$ = redness) and *b* ($-b$ = blueness, $+b$ = yellowness). The *a* and *b* axes are both scaled from -100 to $+$ 100, though higher saturation values are not always achievable in reality. Sometimes, in order to simplify the representation of a colour to two-dimensional space, the *a* and *b* values are transformed to θ, which is the angle between the line joining the point (a_1, b_1) in Judd-Hunter space to the origin and the green/red axis. If saturation is neglected, the colour can then be represented by the two independent quantities *L* and θ, lightness and hue angle (see Fig. 4.6).

Judd proposed that in order to calculate differences between colours in a colour solid, Euclidean geometry could be used such that the 'Euclidean distances between points are proportional to the size or perceptibility, ΔE, of the colour difference between the colours represented by the points'. In the *L*, *a*, *b* colour solid, therefore,

$$\Delta E = [(\Delta L)^2 + (\Delta a)^2 + (\Delta b)^2]^{1/2} \tag{4.1}$$

and across most of the solid a value for ΔE of around unity indicates the maximum acceptable colour difference in the industry.

In the 1960s, the Commission Internationale de l'Éclairage (CIE) decided to try to stem the confusion caused by the presence of the many formulae in use and standardise colour space. They managed to combine the opponent chromatic attributes used by Hunter and Schofield, with the MacAdam cube-root simplification of the earlier Judd polynomial. The result was the publication of CIELAB (or CIEL*a^*b^*) in 1976 (Smith

1997). At this point, the asterisk superscript was introduced in order to differentiate the CIELAB parameters from other similar ones still in use.

Work to create a truly uniform colour space still continues. Following studies by the Colour Measurement Committee (CMC), the CIE94 total colour-difference was defined. Tolerances based on ΔL^*, Δa^* and Δb^* terms give too much area out of visual tolerance but within instrumental tolerance. Using lightness, chroma and hue ($L^*C^*h^*$) attributes allows ellipsoidal tolerance areas to be derived which are much better and easier to describe sensorially than the vague redness-greenness, yellowness-blueness opponent pairs. ΔE^*_{94} is a weighted Euclidean distance using the colour difference components ΔL^*, ΔC^* and ΔH^*

$$\Delta E^*_{94} = \left[\left(\frac{\Delta L^*}{k_L S_L}\right)^2 + \left(\frac{\Delta C^*}{k_C S_C}\right)^2 + \left(\frac{\Delta H^*}{k_H S_H}\right)^2\right]^{\frac{1}{2}} \tag{4.2}$$

where S_L, S_C and S_H are weighting functions used to account for the variation in perceived colour magnitude and k_L, k_C and k_H are parametric functions which correct for sensitivity with variation in experimental conditions. Under reference conditions these factors are assigned a value of unity, whereas in the textile industry, for example, it is common practice to set the lightness parametric function to 2 (CIE 1995).

Use of the CIEL*a*b* and CIE94 is growing in popularity especially when correlation with sensory analysis is required. Relating sensory colour analysis with instrumentally measured data is covered in more depth in Chapter 2 of this book. Of practical note, it is recommended that users comparing data confirm the colour notation and system used – the lightness, and chromaticity co-ordinated used in the Hunter-Judd analysis *L*, *a*, *b* do not correspond exactly with the *L*a*b** of CIELAB. Subsequently, it is essential that full specification including colour system, illuminant type and viewing geometry is quoted when publishing colour measurement data. Furthermore, many web-based applications exist within which one can convert data from one colour system to another, e.g. CIELAB to HunterLab to Munsell Chip Notation (www.colorpro.com/info/tools/convert.htm#TOP).

4.3 Colour measurement methodology

Methods for the instrumental measurement of food colour involve four stages:

1. standardisation of the instrument
2. setting the instrument variables
3. selection, preparation and presentation of the sample to the instrument
4. determination of the colour values.

Bilmeyer (1981) has suggested that deviations from absolute accuracy in colour measurement are caused by, in decreasing order of importance:

- sample preparation and operator variables
- different reference white standards
- variation in colour reference standards
- instrument conditions.

There are many variable factors in each of the four stages of the colour measurement method that can affect the final result. In order to be able to compare results for a

particular food material, these variables should be recognised and efforts made to eliminate, or at least reduce, their effects.

4.3.1 Standardisation of the instrument

Colour standards provide the references against which the colours of materials can be instrumentally compared. They fall into two classes, primary standards and secondary standards. Primary standards are pressed powder tablets of fresh MgO, $BaSO_4$ or halon G–80 (pressed tetrafluoroethylene resin manufactured by the Allied Chemical Corporation) maintained by governmental standards agencies such as the National Bureau of Standards in Washington DC, USA or the National Physical Laboratory in Middlesex, England. These white standards are measured against a theoretical perfect white diffuser by means of an auxiliary sphere to derive an absolute reflectance value. Unfortunately, at the moment there is not a complete consensus between the various standards institutes on the perfect white diffuser values.

Secondary standards can be subdivided into a number of groups. Hunter and Harold (1987) have described seven types of secondary standards in the following terms:

1. Master standards calibrated by reference to primary standards and usually in the form of porcelain or opaque glass panels, ceramic tiles, acrylic painted chips, etc. These are maintained with extreme care, usually at the standards institutes, and are used only to calibrate other secondary standards.
2. Working standards used as a reference for a group of laboratory instrument standards. These are usually ceramic or porcelain tiles.
3. Instrument standards used to calibrate a particular instrument.
4. 'Hitching-post' standards which are used in place of a white standard and which are close in colour to that of the samples to be measured. This technique is used to minimise errors caused by scale-interval defects in the instrument.
5. Specific calibration standards which are permanent specimens or standards used for measurements carried out on impermanent products. To derive these standards the impermanent product is first measured on a reference instrument. The product can then be used to calibrate a second instrument on which the permanent standard is also read. This reading then becomes the assigned values of the permanent standard.
6. Diagnostic standards used for instrument checking or fault finding in inaccurate instruments.
7. Material comparison standards, which are usually samples of the materials under examination and provide limiting or target values.

It can be seen that, in all except the last two cases, the colour values of the standards are traceable, via the working and master standards, back to the primary standard.

Secondary standards should be treated with care. Colour values can be affected by the presence of dirt, grease, moisture, fingerprints or surface scratches on the viewing area. Ceramic standards should be cleaned carefully and regularly by washing in mild anionic detergent, rinsing in hot distilled water and blotting dry with a lint-free paper towel. When not in use, they should be maintained at a constant standard temperature and away from light in a sealed container. Despite even these precautions, it is known that some surfaces can be attacked by atmospheric pollutants or may undergo changes during storage. Painted or printed colour chips should never be cleaned. Freezer storage is often practised to extend the life of colour chips (Huey 1965).

Before a standard is used to calibrate an instrument, the instrument should be thoroughly warmed up and stabilised. Colour standards should be placed under the instrument port in the same manner each time a calibration is carried out, to avoid the effects of any surface directionality in the colour layer of the standard on the calibration values. Because some of the colour standards exhibit thermochromicity, i.e. their colour values change as the temperature changes, it is advised that calibration be carried out with rapidity in order to avoid the instrument lamp raising the temperature of the standard. The subject of thermochromicity of the colour standard has been described by Brimelow (1987) with reference to measurement of tomato colour.

Kent and Smith (1987) reported the results of an inter-laboratory study on measurement of colour standards. They came to the obvious and important conclusion that in order to transfer and compare colour data from one laboratory to another, which is certainly necessary when colour is being used as a buying criterion, the measuring system and the colour standard have to be carefully defined. Variables inherent in the measuring system will be described in the next two sections.

4.3.2 Selection, preparation and presentation of the samples

The sample itself is the key to good colour measurement. In optical terms, samples can be divided into four main classifications: opaque objects, metal-like objects, transparent objects and translucent objects.

With opaque objects, a class into which most food materials fall, light is either directly reflected from the surface as a white highlight (gloss) or absorbed, scattered and diffusely reflected from the surface (perceived colour). Opaque materials do not transmit light. The relative amounts of gloss and colour perceived depend, of course, on the surface properties, but also on the viewing angle. In order to compare colour differences independently of the gloss characteristics, the optimal positions of the light source and the detector (e.g. the eye) relative to the sample surface are 45º and 0º or 0º and 45º. This optical arrangement is replicated in many tristimulus colorimeters and, thus, these instruments measure sample colour and not surface effects. Most spectrophotometers, however, provide data either with or without this specular effect and if managed correctly can provide useful information on product shine or packaging influences.

In the case of metal-like objects, virtually all the light is reflected as gloss and very little is diffused. Very often the reflectance is at a dominant wavelength, which provides colour gloss, rather than at all wavelengths which would provide white highlights. Foods, though they may sometimes have a high gloss component, rarely fall into this class of metal-like materials.

Transparent objects exhibit colour by regular transmission, that is they absorb light and then transmit it preferentially at different wavelengths along the path of the light beam. Some liquid food materials and a few solid food materials are transparent, such as vegetable oils, fruit serums and wines, certain sugar confectionery, fruit jellies, etc.

The class of materials between transparent and opaque, to which many food materials belong, is called translucent. Some of the light falling on a translucent object can pass through it in the direction of the light, but the remainder is diffused within it and emerges in different directions. As a consequence, the size of the measurement aperture can greatly influence the amount of collected light (see Fig. 4.7). The light-scattering properties of hazy or turbid food materials is a complex subject which has been covered in some depth by MacDougall (1982; 1983; 1987). It is clear that with this class of material, instrument/object geometry plays a large part in the colour values obtained.

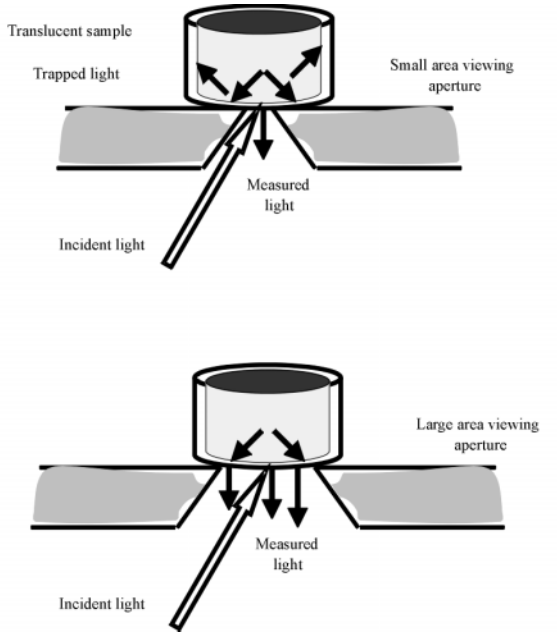


Fig. 4.7 Translucent foods diffuse the incident light resulting in lost measurement. Using a larger area viewing aperture allows this trapped light to be measured.

MacDougall (1987) has illustrated this point using concentrated and diluted orange juices as test samples.

When reflectance measurements are carried out on opaque or translucent food materials, a number of problems exist in the selection and preparation of the samples. The first is that simply by reason of its presentation method, the sample area as presented to the instrument may not be representative of the bulk of the material under test. Most instruments in common use are designed to accommodate samples which are flat and, therefore, reflect light from one plane only. Most food materials are not flat and some degree of compromise has to be made in order to create viewing areas which are. It is often the case that it is advantageous to deliberately change the surface characteristics of the sample in order to enhance colour differences between specimens. André and Pauli (1978), for example, pulverised dehydrated pasta into a powder and then tabletted the powder for colour measurement in order to minimise variation caused by differences in the shapes and surface properties of commercially available products.

A second important problem is that other factors come into play which may affect surface reflection during the process of creating a flat viewing area. These factors include the ambient conditions of temperature and humidity applied during the process; the variables within the actual method used (e.g. the blade sharpness during cutting, the technique of polishing, the pressure applied during pressing); chemical reactions (e.g. browning effects) or physical reactions (e.g. settling effects) occurring at the surface up to and during the measurement event; and the settlement of vapour, dust or film-forming agents on the surface after its formation.

A third problem is the general non-uniformity of the surface colours of foodstuffs. This means that it may be necessary to make repeated measurements in different places on a surface or make measurements of different samples from a bulk in order to achieve statistically significant average values. One ingenious solution to the problem of making

repeated measurements on a surface was suggested by Francis (1952) for apples. By rotating a single fruit rapidly in a small viewing aperture, an average surface colour was obtained.

One further problem is the physical size of the food sample. Some specimens may simply be too small to measure successfully as single items, though recently instruments with very small aperture viewing heads which permit measurements on areas of around 3 mm diameter are available. It is, however, more usual to bulk such samples and then employ large area sample cups and large diameter instrument apertures when making the measurements.

When preparing samples it is important that the preparation procedure is strictly adhered to in order to ensure a high repeatability of measurement. Grinding, milling and mixing may all help to create sample uniformity but at the same time it must not be forgotten that such techniques will undoubtedly influence the sample's light-scattering properties (Giese 2000). In some cases, the physical aspects of the sample can be more influential then the colorant concentration itself. Particle size, size-distribution and component packing may influence the sample optics and hence, the measured colour (Joshi 2000).

Apart from considering all of the above, it is also important to recognise that sample preparation and presentation may be influenced by the desired result of the colour measurement. For example, if the goal is to achieve a good correlation with sensory data, then it is essential that the instrumentally measured sample should be in the same state as that seen by the visual panel. If colour measurement is to be used as a buying criterion the trial sample must be presented as was the standard used to set the tolerance limits. Also, if a relationship between colour and some other physical parameter is to be made, e.g. particle-size or water-activity, both samples should undergo the same preparation for each test. Some general rules for the preparation of different types of food materials for reflectance colour measurement are discussed below.

Opaque powders, granules and flakes

It is usual to make multiple measurements on such materials and average the results. Bulk amounts are normally sieved prior to subsampling in order to reduce errors due to size variation in the particles. Hunter and Harold (1987) have clearly illustrated that the larger the particle size, the less the light is scattered and therefore the more coloured the sample appears. Smaller particles scatter and reflect more incident light and the sample therefore appears lighter (see Fig. 4.8). Many of the errors encountered in comparing the colours of powders and granules relate to particle size effects.

One of three methods of sample preparation of powders can be utilised. Samples can be poured into cups with optically flat and transparent bases and measurements made through the base. Alternatively, samples can be poured into a container with sides of equal height and the top can be scraped flat with a straight edge. Measurements are then made from above. Thirdly, a powder compactor can be used. This is a device which can create a disk specimen from a sample of known weight and bulk density by applying a standardised force. In this case measurements can be made from above or below.

Opaque particulate or lumpy materials

Materials from 5 mm to 20 mm in size can be very difficult to measure with good repeatability, particularly if they are non-regular in shape. As has been mentioned, several instruments have recently appeared on the market with small diameter viewing heads, such as the Minolta CR 121 or the MacBeth ColorEye XTH. Normally, however,

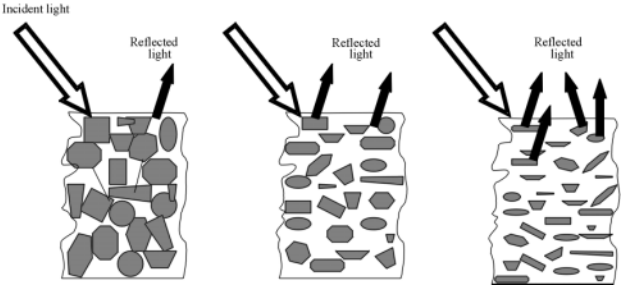


Figure 4.8 Decreasing particle size increases light scattering, causing the sample to appear lighter in colour.

it is the practice to place particulate or lumpy materials into a large container with an optically flat and clear base, trying to ensure that the packing is as solid as possible. Measurements are then usually made from below.

An alternative method of presentation has been evolved which is useful in situations when repeated measurements have to be made on solid materials which have a high oil or fat content. These materials may smear or coat the surfaces of a glass cell. The method utilises clear PVC film (e.g. Saran Wrap). The film is stretched across the viewing port of the colorimeter and the sample is placed on the film. Measurement is made from below. After each measurement, the specimen can be quickly wrapped up in the film and discarded, thus eliminating the need for frequent washing of valuable glass sample cups (Mabon 1989).

Large area solid foods

When colour measurements are made on solid opaque foods, such as whole fruits or thick slices of red meat, the specimen field should be flat and uniform. If a natural flat surface is not available, then it is sometimes acceptable to form such a surface by cutting or pressing. The latter can be achieved by sandwiching the sample between two plates of glass, one of which should be optically flat and transparent. Viewing is carried out through the transparent plate. If the food does not lend itself to cutting or pressing, then one other possibility is to rotate the food object in front of the viewing port of the instrument (Francis 1952). Solid foods often show directionality. When the directionality is obvious, the specimen should be measured in the same orientation for each test. When the directionality is haphazard or undefined, one useful technique is to turn the sample through 90º between the two readings and average the results.

Many solid foods are translucent, for example, cheeses, butters, fatty meats and high water content fruits and vegetables. The effects of translucency on colour readings are large, particularly when such materials are being measured in thin slice form (for example, bacon, other sliced meats and sliced fruits). It is essential to standardise the preparation procedures for such samples. Normally a standard white backing plate would be employed and the sample thickness would also be standardised. Specimens are often viewed through an optically clean and flat plate placed on the surface of the solid. Some of the effects of the preparative variables on the colour values of sliced meats have been discussed by MacDougall (1982). A few food solids are transparent and they can be measured by transmittance procedures. Specimens of known thickness are held against the viewing port of sphere-type colorimeters/spectrophotometers for measurement.

Pastes and slurries

Food pastes and slurries are normally translucent materials. It is usual with such materials to introduce them into cells with optically clear and flat bases, to such a depth that all the light is trapped and diffused within. Measurements are made from below. In some instances, however, particularly when the aim is to compare lightness values in dark samples or when a standard background is desirable, a special cell equipped with a ring insert of known height is employed. The sample slurry is poured into the cell to fill it to above the ring and a dish with a white diffusing surface is pushed down to nest on the ring. This fixes the light path length of the sample when it is viewed from below.

Often it is advisable to cover the specimen cell with a black lined box in order to prevent extraneous room light from passing into the specimen. Care should be taken when introducing pastes and slurries into test cells that no air is entrapped in the sample. When pastes and slurries are prepared by mincing followed by diluting with water, the mincing conditions must be standardised as must the dilution factor. Samples should be well stirred and then measured quickly to reduce the effects of sedimentation and oxidation. Finally, the thermochromic properties of certain food materials should not be forgotten, and sample temperatures should be standardised.

The effects of some of these preparative factors – dilution, test temperature and time, presence or absence of a black cover – on the colour values of tomato pastes have been discussed by Brimelow (1987). As already noted, MacDougall (1983) has illustrated the effects of the dilution factor on colour values in the case of orange juices.

Liquids

Liquids can be translucent (milk, syrups, fruit juices, egg yolk) or transparent (wines, vinegars). Translucent liquids are measured in the same way as translucent slurries and pastes, and once again choosing the correct optical path-length can determine the validity of the measured result. Transparent liquids are measured by pouring into optically clear fixed path length cells with parallel sides and then taking transmission readings in sphere instruments. Deeply coloured liquids such as dark treacles are measured in short path length cells of the order of 2 mm, whilst lightly coloured liquids such as white wines are measured in long path length cells up to 50 mm. Alternatively, using a cell backed with a white tile may permit colour measurements by transreflectance. In such cases the effective path-length becomes twice that of the cell since the illuminant light travels through the sample, reflects from the white tile or mirror backing, and then traverses again through the sample before being measured.

4.3.3 Setting the instrumental variables

The instrumental variables include the choice of light source, the measuring geometry and the relative sizes of the specimen area and the illuminated area. When any colour measurements are made, it is necessary to specify the illuminant used as part of the viewing conditions. In 1931 the CIE established various standard illuminants which have characteristics close to natural light sources, but which also can be reproduced easily for use in the laboratory or in instrumentation. Illuminant A represents light close to that given by a tungsten filament lamp, which might for example be operating during subjective assessments of colour in the home. Illuminant B represents direct sunlight, whereas illuminant C represents average daylight from the total sky, which again might be present during assessments in the home. More recently, the CIE (1979) proposed a D series of illuminants which more closely represent daylight than do B or C and also

incorporate an ultraviolet range. The D series correlates with various daylight colour temperatures; the most commonly used is the D65 illuminant, approximating to a colour temperature of 6500 K.

It should always be borne in mind, however, that if a subjective assessment of the colour of a particular food is normally carried out under, say, a tungsten filament light or under direct sunlight, then it may be preferable to carry out the instrumental measurement under similar lighting conditions. It should be noted also that a pair of samples which colour-match another under one illuminant may not match under a different illuminant. This phenomenon is known as metamerism and is of great importance to those in the textile and paint industries.

The choice of the remaining instrumental factors, such as measuring geometry and the relative sizes of the specimen viewing area and the illuminated areas, are very much influenced by the optical classification of the specimen. In order to determine the colour values of a product, therefore, it is necessary to decide whether the prepared sample is opaque, translucent or transparent. Transparency is relatively simple to assess; when a clear image of a light source can be seen through the sample it can essentially be considered as transparent. Translucency is more difficult to determine as the dividing line between opacity and some translucency is imprecise. A simple test is to measure the specimen with maximum area of illumination and maximum viewing aperture and then to repeat the measurement using the same viewing aperture but a reduced area of illumination. If there is a large increase in the lightness reading (the L^* value in the CIELAB scale), then the product is translucent. Another good indicator is to measure the sample over firstly a black then a white background. If these two measurements are the same, then the sample can be considered to have an infinite optical thickness.

The viewing geometry must also be specified as part of the instrumental conditions. It is generally agreed that with food samples most applications can be covered without much compromise using 0º/45º, 45º/0º (source/detector), or specular component excluded (SCE) viewing geometry. This is the preferred geometry when examining the reflected colour of translucent samples. It is also satisfactory for opaque samples, provided the samples being compared are of equal texture or gloss. Even samples with different texture or gloss characteristics can be compared using this geometry, provided that appearance differences rather than simply colour differences are being assessed. SCE geometry should be used for transparent samples or for measuring transmittance through translucent samples.

One source of error arising from the viewing geometry occurs through failing to define whether specular reflectance is included or not. With 0º/45º or 45º/0º geometries, very little specular reflectance is included in the total reflected light detected, even with glossy but non-smooth samples. However, with spherical (d/8º) geometry, specular reflectance will be present in the reflected light from glossy samples and the decision has to be made to include it or exclude it.

The choice of illuminated area size and aperture size is very dependent on the nature and the optical class of the specimen. Small area illumination is used on small samples, on samples having only small areas of flatness and on highly translucent and uniform samples. In the latter case, Hunter and Christie (1978) have recommended that the small area illumination is used together with large aperture viewing in order to capture that proportion of the light scattered by the sample and then returned to the detector from outside the illuminated area (see Fig. 4.7). As has been mentioned already, particulate solids are best measured using large area illumination and large viewing apertures.

4.3.4 Determination of colour values

The last stage in obtaining a colour measurement is the conversion of the raw signal to colour values in one or other of the colour scales. In modern instrumentation, software packages are capable of carrying out these conversions with speed and accuracy, and can even make calculations to predict comparative colour results for different light sources. Since calculations are based on reference white values, it is a point of concern that there are still differences between national standards for the perfect white diffuser.

4.4 Colour measurement of typical food materials

In this section, some typical applications of food colour measurement will be described by reference to practical work in this area. The work will be treated under the headings utilised earlier in the chapter: powders, granules and flakes; particulate and lumpy materials; large area solid foods; pastes and slurries; and liquids. For further details on the preparation procedures, the sample temperature, the sample presentation, the instrument illumination and aperture diameters, the colour standard and the colour scale employed in each case, the reader may refer to Tables 4.1 to 4.5, respectively. The instruments utilised in each case are briefly summarised in Table 4.6 on page 111.

4.4.1 Powders, granules and flakes (Table 4.1)

Johnston *et al.* (1980) reported that several researchers had compared colorimetric values with visual judgements for durum semolina, but that no one had compared colorimetric data with extracted pigment content. They found that semolina reflectance *b* values were highly correlated with pigment content as determined by pigment extraction and estimation using the standard AACC water-saturated butanol method, particularly when the objective was to segregate highly pigmented semolinas from low pigmented ones. The authors pointed out that reflectance colour measurement in these types of applications could be a genuine timesaving alternative to the lengthy pigment extraction technique.

Seakins (1971) developed a method to measure the colour of casein by immersing it in a liquid of similar refractive index in order to overcome the effects of particle size. Reflectance measurement of dry sample preparations resulted in poor separation of colour grades because reflectance increased as particle size decreased. However, when transmittance values were measured on the liquid suspension, the colour grades were clearly differentiated. In order to minimise grind size effects in ground coffees of different roasts, Little and MacKinney (1956) standardised on a very fine grind with maximum lightness value.

Berset and Caniaux (1983), working with parsley, found that measurements on the ground leaves gave better reproducibility than measurements on the whole leaves. The authors surmised that grinding the leaves released from interior leaf cells chlorophyll pigments which were of a different type and more highly pigmented than those in the surface cells. It was shown that the hue function correlated highly with extracted chlorophyll *a* and *b* content. These workers also suggested that their colorimetric procedures could be utilised in the quality control of parsley drying and storage processes.

In order to overcome the problem of colour variation across potato crisps, Habib and Brown (1956) ground them down to a uniform colour and then introduced the size-

Table 4.1 Colour measurement of powders, granules and flakes

Reference	Sample	Preparation	Temperature (°C)	Presentation (viewed from above/ below)	Illumination/ aperture diameters (mm)	Standard	Instrument[a]	Measurement[b]
Johnston *et al*. (1980)	Durum semolina	Equilibrate to 15% H_2O, mill to semolina flour	NM	20 g samples	NM	NM	A(1)	*b*
Seakins (1971)	Casein	Sieve to 6 different particle ranges. Suspend samples in benzene	NM	0.5 cm light path	T[c]	Transmittance standard at 455 run	B	%T at 455 mn,
Little and MacKinney (1956)	Coffee beans at different roasts	Beans ground in commercial coffee grinder, sieved to 5 size ranges	NM	Ground coffee in clear glass cell	NM	White tile, neutral grey tile	C	*Y*
Berset and Caniaux (1983)	Dried parsley leaves	Leaves ground 1 min in ball grinder, sieved to 80–250 μm particles	NM	NM	NM	NM	D	*L*, a*, b* X, Y, Z, X, Y*
Habib and Brown (1956)	Potato crisps	Crisps size reduced by milling 5 min in Oster mill	NM	Viewing cell 89 mm high by 51 mm diameter. Sample depth of 76 mm (below)	NM/51	Ivory tile	E	*L, a, b*
Grieder (1989)	Herbs	Mix 5 g herbs with 5 g sand using automated mortar mill	20	Black metal cell with glass base, 55 mm diameter by 60 nun deep (below)	51/55	White tile	A	*L, a, b,* ΔE, ΔC
Hans and Floros (1998)	Potassium sorbate powder	Ground to a fine powder then baked	0	Glass cell	NM	NM	A(6)	*L,a,b*

[a]For instrument codes see Table 4.6; [b]For colour scales see Chapter 3, Table 3.1.; [c]Transmission cell; NM: not mentioned in the reference.

reduced samples into large volume sample cells using a defined tap-down procedure. Colour differences among four different potato varieties and different storage/conditioning environments were compared over storage time. It was shown by carrying out multiple correlations against Hunter *L* values that reducing sugars and free amino nitrogen were important in colour formation in the final chips.

A novel approach to the preparation of flaky materials for colour measurement has been suggested by Grieder (1989). Working with herbs, he utilised a pulverisation procedure with sand. This aided the dispersion and trapping of oils, pigments and moisture expressed from the interior cells of the test materials during the grinding process.

An example of powder colour measurement for process validation is provided by Han and Floros (1998). They modelled the colour change of potassium sorbate during a series of time-temperature varying trials. Statistically significant changes in both *L* and *a* values suggest that a potassium sorbate based system may prove a good indicator for processes, such as dry-heat sterilisation, which involve high-temperature heating.

4.4.2 Particulate and lumpy solids (Table 4.2)

The subject of colour measurement of dehydrated potato products, such as sliced or diced potato, has been well covered by Bergthaller and his team. In the case of diced potato, for example, Bergthaller *et al.* (1983) packed the cubes into a dull black aluminium cell with an optically clear glass base. The *L*, *a* and *b* values of the potato cubes were compared with sensory data and a high correlation between the two sets of data was obtained.

A number of different food materials were examined by Baardseth *et al.* (1988) as part of a comparative exercise involving two colour measuring instruments, the Minolta and the HunterLab D25P. One of the subject materials was salami-type dry sausage, which was prepared and sliced thinly for colour measurement. The sausage slices had a varied appearance, with visible fat particles of 1 to 3 mm diameter present on the surface. Correlation coefficients between instruments of over 0.9 were obtained, with much of the residual non-predictability being related to the inconsistencies in the slice surfaces.

Lee *et al.* (1988) investigated frozen cut green beans given various blanch treatments. The beans were boiled, drained and placed in a 10 cm square acrylic measuring cell. Five replicates per treatment were measured. Results revealed that significant differences were found in the Hunter *a* values, with beans blanched at temperatures above 82ºC showing a greener colour than beans blanched at temperatures below 82ºC.

In order to eliminate problems associated with the differences in surface properties and shapes of commercially available dehydrated vermicelli, André and Pauli (1978) ground the samples down to powders, then formed tablets from the powders using a hydraulic press. The 40 mm diameter by 10 mm thick tablets could be measured directly on the head of the colorimeter. Good correlations were found between colour co-ordinate values and the β-carotene content derived from the egg component of the pastas. Pastas made from different flours could also be differentiated in terms of colour values.

A non-sacrificial index of peanut kernel maturity has been derived by Miller and Burns (1971) using a colorimeter procedure. Individual cotyledons were placed on the port of a narrow light beam instrument. The half hulls thus acted as miniature hemispherical light-scattering chambers. Ageing of peanuts is accompanied by darkening of the vein cells near the internal hull surface, and this change was measured by a change in the lightness scale readings obtained.

Hazelnuts have been evaluated by Ozdemir and Devres (2000). They measured the colour after roasting of both the intact kernel and a ground hazelnut powder. Significant

Table 4.2 Colour measurement of particulate and lumpy solids

Reference	Sample	Preparation	Temperature (°C)	Presentation (viewed from above/ below)	Illumination/ aperture diameters (mm)	Standard	Instrument[a]	Measurement[b]
Bergthaller *et al*. (1983)	Potato cubes, dried	Commercially available cubes packed into cell to depth of 60 mm	NM	Dull black Al cell 60 mm internal diameter, height 60 mm, with optical glass base 1.2 mm. thick (below)	50/NM	White enamel tile	G	y, *L, a, b*
Baardseth *et al*. (1988)	Dry sausage, salami type	Slices, 3mm thick	NM	No cover used, measured against black background	50/50 25/32	White tile	H(1) A(4)	*L*, a*, b**
Lee *et al*. (1988)	Frozen green beans	Boiled 5 min. from frozen. Drained 1 min.	NM	10 cm square acrylic cell	Large/NM	White tile	A	*L, a, b*
André and Pauli (1978)	Dried vermicelli (of various shapes)	Pasta ground, sieved through 0.28 mesh, 10 g of powder formed to a tablet 40 mm diameter by 10 mm thick	NM	Faces of tablets measured directly on sensor port	NM/10	White enamel tile	G	*X, Y, Z x, y*
Miller and Burns (1971)	Peanut hulls (internal colour)	Peanuts harvested, shelled, and sorted subjectively into mature or immature categories	NM	Peanut hull halves directly on narrow beam sensor port	Narrow/NM	Porcelain coloured tile	I	Rd[c], *a, b*
Authors' laboratory (1988)	Fresh pasta, angel hair and tortellini	Pasta portions packed into 89 mm diameter cell to cover optical glass base, flat against glass when possible	20/22	Reflectance through glass base. Black cover on cell (below)	50/89	White tile and black glass	I(3)	*L, a, b*
Ozdemir and Devres (2000)	Hazelnuts	Whole roasted kernel and ground roasted powder	NM	Direct onto kernel	NM	NM	H(3)	*L, a, b*

[a] For instrument codes see Table 4.6; [b]For colour scales see Chapter 3, Table 3. ; [c]Luminosity value for some older instruments. Rd = *Y; NM: not mentioned in the reference.*

changes occurred in both *L* and *a* measured values with differences between whole and ground nut results suggesting that roasting level should not be monitored by considering either of these uniquely. By fitting the data to an Arrhenius type model, the authors established a generalised model describing colour changes as a function of temperature and time.

Fresh pasta colour has been measured at the laboratory of one of the authors' (CB) (1988), the subject materials being spinach and egg varieties of ravioli, fettuccine, angel hair and tortellini. Samples were provided packed under gas atmosphere. It was found that food substances of this type required immediate measurement after removal from the package since surface colour changes started to occur as soon as the specimens were exposed to air. The effect was most noticeable with thinly cut angel hair. Samples were placed quickly into large area glass cells and measured at a large aperture instrument port within 30 seconds of removal from the package.

4.4.3 Large area solid foods (Table 4.3)

Sapers and Douglas (1987) measured enzymatic browning effects at cut surfaces of pears and apples, by cutting 22 mm diameter plugs from the samples using a cork borer and then placing the plugs on a glass cell over the instrument port. Browning could then be followed with time by measuring the *L* and *a* values. A similar preparation technique has been utilised by Konstance *et al.* (1988) on beef chuck. In this case, samples were taken from 2.54 cm thick slices of chuck using a cork borer. The cylinders were then inserted in a previously bored rubber stopper placed on a glass cup. Components of the chuck meat such as lean, fat, bone and ligament could be separated in terms of ΔE and colour values.

Though many of these examples deal with the subject of tristimulus colorimetry, it is important to recognise that reflectance spectrophotometry is also a useful technique for the measurement of colour changes in certain foods and one that is growing in popularity as the costs of spectrophotometers rival those of colorimeters. One example is the colour change from red to brown occurring in meat due to oxidative/reductive reactions involving myoglobin. This colour change has a direct effect on the marketability of cut meat pieces. Eagerman *et al.* (1978) have followed the colour change with time in beef semimembranosus muscle. Approximately 1 cm thick slices were placed against the port of a spectrocolorimeter and the reflectance spectra measured between 600 and 650 nm. The formation of a depression in the region of the spectral curve around 632 nm is an indicator of brown metmyoglobin pigment production.

The use of transmittance colorimetry on solid materials is illustrated by the work of Freedman and Francis (1984) who have measured the colour of jellies. The jellies were heated sufficiently to pour into 2 cm light path cells. Lightness and hue angle were used to follow the effects of storage time, with or without the presence of ascorbic acid in the formulations.

Measurements on whole fruits have been carried out by Nagle *et al.* (1979) and Reeves (1987) for capsicums, Shewfelt *et al.* (1989) for fresh tomatoes and Lin *et al.* (1989) for apples. In all cases the fruits were measured directly on the instruments, taking a number of readings around the circumference of each specimen. High correlations were obtained between colour values of the whole fruit and extracted xanthophyll pigment in red peppers. Shewfelt *et al.* (1989) showed that for whole tomatoes there was a significant difference between colour values (colour difference DE, and difference in chroma $[a^2 + b^2]^{1/2}$ due to cultivar, storage treatment and evaluation period. Simulated warehouse storage conditions were found to delay ripening colour changes.

Table 4.3 Colour measurement of large solids

Reference	Sample	Preparation	Temperature (°C)	Presentation (viewed from above/ below)	Illumination/ aperture diameters (mm)	Standard	Instrument[a]	Measurement[b]
Konstance *et al*. (1988)	Beef chuck	Sliced 2.54 cm thick	25	Glass cup (below)	NM/19	White plate	I(1)	ΔE, ΔC, YI[c]
Eagerman *et al*. (1978)	Beef semimembranosus muscle	Sliced 12 × 8 × 1 cm	0	NM	NM	NM	K	R (632, 614 nm)
Freedman and Francis (1984)	Fruit jellies	Juice-sucrose-pectin jellies	NM	4 × 5 cm transmission cell, 2 cm light path	T	Distilled water	A(5)	L, θ
Nagle *et al*. (1979); Reeves (1987)	Whole red peppers	Blanched whole fruit	NM	Measured around the shoulder of the fruit	NM	Colour plate	I(4)	L, θ
Shewfelt *et al*. (1989)	Fresh tomatoes	Whole, measured 'as is'	NM	Measured at 8 equidistant locations on surface of fruit	NM	Pink tile	I(6)	L, a, b θ, chroma
Lin *et al*. (1989)	Apples, fresh	Whole, measured 'as is'	NM	Measured at 8 equidistant locations on surface of fruit	NM	Pink tile	I(6)	L, a, b, θ
Sapers and Douglas (1987)	Raw apples and pears	Plug (22 mm diameter) removed from halved fruit. Transverse cut across plug just prior to measurement	20	Fresh transverse surface of plug centred over instrument aperture	Large/19	White tile	I(1)	L, a, b, X, Y, Z
Baardseth *et al*. (1988)	Potatoes, raw	Cut lengthwise, covered with polyethylene film	NM	Readings taken immediately through poly film	8/8 50/50 25/32	White tile	H(2) H(I) A(4)	L^*
Baardseth *et al*. (1988)	Raw cauliflower	Cauliflower was measured 'as is' directly at port	NM	No cover used	50/50 25/32	White tile White tile	H(1) A(4)	L^*, a^*, b^* L^*, a^*, b^*
Hetherington *et al*. (1990)	Fresh tomatoes	Directly on tomato equator	12		NM/20	NM	M	XYZ, L^*, a^*, b^*

[a]For instrument codes see Table 4.6 ; [b]For colour scales see Chapter 3, Table 3.1; [c]Yellowness index YI = 100/Y (1.277 X- 1.06Z) NM: not mentioned in the reference. T: transmission cell.

Significant red colour loss and increase in browning were detected in modified atmosphere packaged apples by Lin *et al.* (1989). The unpackaged apples had more yellow colour loss than shrink-wrapped apples. The authors extracted three pigments from apple skin and showed that the highest correlation was obtained between cyanidin-3-galactoside and Hunter *a* values.

A useful technique for raw fresh cut potatoes has been employed by Baardseth *et al.* (1988). The pieces were wrapped in polyethylene film to retard browning and surface drying before placing on the instrument port; slices were then measured immediately. The L^* values were compared between two instruments. Significant correlations were obtained between results from the Minolta large measuring area instrument (CR-110) and the HunterLab large measuring area instrument, as well as between the small area Minolta (CR-100) and the HunterLab instrument.

Raw cauliflower was also measured by Baardseth *et al.* (1988). Measurements were made directly without extensive sample preparation using both large area measuring apertures on the Minolta (CR-110) and HunterLab units for comparison. The colour values obtained for cauliflower were found to overlap with mashed potato measurements. The authors suggest that data from these two food commodities could be combined to develop a regression equation that would allow a researcher to predict lightness and yellowness from the onset of treatment.

Hetherington *et al* (1990) made tristimulus colour measurements on whole tomatoes. Using a^* data they obtained significant correlation with both a mechanical measure of ripeness (puncture test) and with sensory observations associated with colour changes from green to red. By sectioning the fruit samples (2 mm thickness) and then measuring reflectance data over white and black backgrounds, absorption and scattering coefficients based on Kubelka-Munk theory were also obtained. These provided the necessary link, when combined with colorimetric data, to quantify the aspect of translucency. Ripeness, as used as a harvesting/buying criterion, requires both visual and tactile assessment and so by combining colour, mechanical and sensory testing this study clearly shows the potential for development of on-vine quality control instrumentation.

4.4.4 Pastes and slurries (Table 4.4)

This group of materials offers the most problems in colour measurement terms, because of the influence of translucency and light trapping effects on the colour results. Baardseth *et al.* (1988) reported that the prediction index for yam colour measurements was the least precise in the red/green parameter than for all the other foods evaluated in the paper, such as mashed potato, raw potato and raw cauliflower. The authors suggested that homogeneity was a key factor in obtaining the most precise prediction index, and that in the case of yams, therefore, variations in the fruit pieces may have been a factor in the loss of precision.

Nagle *et al.* (1979) homogenised pepper pods, the seeds and stems having been removed, prior to introduction of the sample into a cup provided with a white backing. Care was taken to allow the purées to settle; they were then vibrated to remove air bubbles. Good correlations were obtained between colour values and both xanthophyll and total pigment concentrations.

The influence of blueberry content on topping mix colour has been investigated by Francis (1985). Blueberries were added to topping formulations as a purée. The formulations were then pasteurised and cooled before conventional measurement in glass cells allowing ‘infinite’ sample thickness.

Table 4.4 Colour measurement of pastes, purées and slurries

Reference	Sample	Preparation	Temperature (°C)	Presentation (viewed from above/ below)	Illumination/ aperture diameters (mm)	Standard	Instrument[a]	Measurement[b]
Baardseth *et al*. (1988)	Mashed potatoes and yams	Commercially available samples	NM	Plexiglass cell, 10cm cube with 3 mm quartz glass front	50/50 25/32	White tile	H(1) A(4)	L^*, a^*, b^*
Nagle *et al*. (1979); Reeves (1987)	Red pepper purée	Milled whole fruit	NM	Sample cell with white top plate (below)	NM	Colour plate	I(4)	L, a, b
Francis (1985)	Blueberry toppings	Puréed blueberries incorporated in topping mix with 15% or 30% fruit content. Formulations pasteurized	NM	3 cm depth in 6 cm diameter cell (below)	NM	White tile	I(1)	L, θ
Clydesdale and Francis (1969)	Spinach purée	Milled spinach processed to F_o 4.9	NM	NM	NM	Grey tile, pressed Ba_2SO_4	A C L	X, Y, Z L, a, b G, R, B
Huang *et al*. (1970)	Squash purée	Commercially available samples	NM	6.4 cm diameter cells, sample thickness between 2 and 8 mm, backed by black or white papers	14/47 47/47	Pressed Ba_2SO_4 pink tile A	L	X, Y, Z
Silva *et al*. (1989)	Sweet potato mash	Various potato sizes were canned or steamed, then frozen or canned	NM	NM	NM	Orange tile	A	L, a, b θ, ΔC
Bosset *et al*. (1986)	Yoghurt	Commercially available samples	NM	500 ml measuring cell, yoghurts stirred before measurement	NM	NM	F(1)	L, a, b
Brimelow (1987)	Tomato paste	Dilute to 12% TSS[c]	20	50 mm depth in 62.5 mm diameter cell (below). Black cover	51/51	Red tile	A(2)	a/b

[a]For instrument codes see Table 4.6; [b]For colour scales see Chapter 3, Table 3.1;[c]TSS: total soluble solids (%) measured on the sugar scale; NM: not mentioned in the reference.

Clydesdale and Francis (1969) measured the colour of spinach purées formed by blanching fresh spinach, comminuting, deaerating and finally processing to F_0 4.9 in glass jars (F_0 being the time-temperature sterilisation value, related to minutes at 121ºC). Measurements were made on a tristimulus colorimeter 'hitched' to a grey colour tile standard. Squash purées have been measured by Huang *et al.* (1970) using reflectance techniques in conjunction with the Kubelka-Munk concept. In this concept, reflectance measurements of thin layers of material are made over both white and black backgrounds. A reflectance value at infinite thickness can be calculated as well as ratio values for light scattering and absorption. In the work on squash purées, a tristimulus colorimeter was employed previously 'hitched' to a pink standard. Good correlations were achieved between visual ratings and colour values, and between visual ratings and Kubelka-Munk absorption/scattering ratios.

Silva *et al.* (1989) examined sweet potatoes prepared by various methods. All samples were mashed or puréed for chemical and physical analyses. The authors utilised an orange colour tile for 'hitching' standardisation of a Hunter D25 instrument. Colour differentiation of the process variables in the *L*, *a*, hue angle and saturation or chroma measurements could be used, the authors suggested, in predicting the quality of these types of products. The lower *L*, *a* and saturation values of canned samples versus their frozen equivalents were indicative of the more extensive heating which the canned products received.

In an extensive study to evaluate the effects of packaging materials, light and storage on solid, whole milk yoghurt, Bosset *et al.* (1986) measured colour changes in the yoghurt in a specially designed 500 ml capacity cell for the MacBeth Spectrophotocolorimeter. Their results indicated that changes in the *L*, *a* and *b* colour values could be related to exposure to light through the various packages. In particular, the authors reported that Hunter *a*, *b* values were sensitive indicators of product changes due to photodegradation.

Finally, an exhaustive study of some of the variables affecting the measurement of tomato paste colour has been carried out by Brimelow (1987). The effects of such factors as paste dilution, test duration, instrument aperture size and illumination area, sample temperature, standardisation hitching-post tile temperature, and the presence or absence of extraneous light were all examined. A standard colour measurement procedure was advocated for this important food commodity.

4.4.5 Liquids (Table 4.5)

Many clear or lightly translucent liquids are measured by transmission techniques, as already discussed. Thus Francis (1985) measured the colour of blueberry beverages in 2cm cells, Calvi and Francis (1978) measured grape beverages in 1 cm cells, Skrede *et al.* (1985) measured blackcurrant syrups in 1 cm cells after diluting to 12º Brix, and Pilando *et al.* (1985) measured strawberry wine in 1 cm cells.

The study of Skrede *et al.* (1985) illustrates how colour measurement can be used to predict the shelf-life of a particular food material. Commercial blackcurrant syrups were packed in various packaging materials with different barrier properties. Storage lives varied between 2 months in high density polyethylene, 12 months in polyester and 18 months in glass. Combinations of Hunter *L*, *a* and *b* values were found to give acceptable predictive abilities for visual colour.

Francis (1985) found in the blueberry beverages that an increase of blueberry content from 5 to 12 per cent did not result in a corresponding increase in visual colour. Since the

Table 4.5 Colour measurement of liquids

Reference	Sample	Preparation	Temperature (ºC)	Presentation (viewed from above/ below)	Illumination/ aperture diameters (mm)	Standard	Instrument[a]	Measurement[b]
Francis (1985)	Blueberry beverages	Puréed blueberries incorporated in beverage mix with 5% and 12% fruit content. Formulation pasteurized	NM	2 cm transmission cell	T	Distilled water	I(2)	*L*, θ
Francis and Clydesdale (1972)	Egg Yolk	Homogenize	NM	20 mm depth glass (below)	10/NM	NM	A	*a*, *b*
Calvi and Francis (1978)	Carbonated grape beverage	Grape anthocyanins incorporated in carbonated beverage formulation	NM	1 cm transmission cell	T	Distilled water	I(5)	*a*/*L*
Skrede *et al*. (1985)	Blackcurrant syrups	Dilution of the syrups 1 to 4 with H_2O	NM	1 cm transmission cell	T	Water	J	*L*, *a*, *b*
Pilando *et al*. (1985)	Strawberry wine	Thawed berries, sugar and yeast fermented to wine, then racked and bottled with SO_2 wine stored in the dark	NM	Transmission in a 1.0 × 5.0 × 5.0 cm lucite cell, with gloss excluded	T	White tile	A(5)	*L*, *a*, *b*
Desarzens *et al*. (1983)	Liquid milk products	Cell filled with ca 500 ml of liquid	NM	Reflectance in 500 ml sample cell	NM	White tile	F(1)	*L*, *a*, *b*
Cole and Roberts (1997)	Gelatine	Molten solutions	40	50ml cuvette with a white tile cover	NM	Instrument white and black tiles	N	*L*, *a*, *b*
Authors' laboratory (1999)	Liquid milk	UHT milk of varying fat content	25	1,2, 5, 10mm cuvette white and black backing	5×10	White tile	F(2)	*L*a*b**, ΔE*
MacDougall (1987)	Evaporated milk, Orange juice, Tomato paste	Dilution with distilled water to obtain various specified solids contents, giving opaque to translucent solutions	NM	Measuring cell containing over 5 cm depth. Black cover	50 down to 10/50 down to 5	White tile	A(2)	*L*, *a*, *b*, *y*

[a]For instrument codes see Table 4.6; [b]For colour scales see Chapter 3, Table 3.1; NM: not mentioned in the reference; T: transmission cell.

Table 4.6 Colour measurement instruments

Code for Tables 4.1–4.5	Instrument	Geometry	Illuminant source
A	HunterLab Color Difference Meter D25	45°/0°C	C
(1)	D25A	45°/0°	C
(2)	D25M	Circumferential 45°/0°	C
(3)	DL25	Circumferential 45°/0°	NM
(4)	D25P	Integrating sphere 0°/*d*	C
(5)	D25 with transmission	Transmission	NM
(6)	D25-PC2	45°/0°	NM
B	Hilger Biochem	Transmission	NM
C	Colormaster Differential Colorimeter	Tristimulus reflectance	C
D	DU COLOR Color Difference Meter (Neotec)	Circumferential illumination	NM
E	Hunter Color and Color Difference Meter	Tristimulus reflectance	NM
F	Macbeth Spectrophotometer		
(1)	MS-2000	Integrating sphere 0°/*d*	C
(2)	ColorEye 7000	Integrating sphere *d*/8°	D65
G	MOMCOLOR D (MOM Hungarian Optical Works)	Tristimulus colorimeter	C
H	Minolta Chroma Meter		
(1)	CR-110	*d*/0°	C
(2)	CR-100	*d*/0°	C
(3)	II	Tristimulus colorimeter	NM
I	Gardner Automatic Color and Color Difference Meter	Tristimulus colorimeter	NM
(1)	XL-23	Tristimulus colorimeter	C
(2)	XL-23 with transmission	Transmission	NM
(3)	Colorguard 2000/45 sensor	Circumferential 45°/0°	C
(4)	XL-10A	NM	C
(5)	XL-10 with transmission	Transmission	NM
(6)	XL-845	Tristimulus colorimeter	NM
	Shimadzu 300-UV	Transmission	C
K	Bausch & Lamb Spectronic 20 with reflectance attachment	NM	NM
L	GE Recording Spectrophotometer	NM	NM
M	Instrumental Color System Micromatch 2000	*d*/8°	D65
N	BYK-Gardner Color-View Spectrophotometer	NM	NM

NM: not mentioned in the reference.

berries are highly pigmented, he surmised that a much greater increase in pigment would be required to achieve an observable change in the beverage colour. Calvi and Francis (1978) were able to correlate the anthocyanins content of the extracted concord grape filter trim with the *a*/*L* index of grape beverages at various storage temperatures. Pilando *et al.* (1985) found that both anthocyanin content and Hunter *L* values were highly correlated with sensory panel judgements when assessing the colour quality of strawberry wine.

Translucent liquids have been measured by Francis and Clydesdale (1972) (egg yolks), Desarzens *et al.* (1983) (liquid milk products), MacDougall (1987) (evaporated milk, orange juice) and others. These materials are measured by conventional reflectance techniques, ensuring usually that samples are at 'infinite' thickness. Work in the authors' laboratory (JP) (1999) has used a path-length variability technique to get a quantitative measure of translucency in milk samples of varying fat content. By using cuvettes of systematically increasing thickness (1 mm–10 mm) and measuring over black and white backgrounds, samples could be characterised with the results comparing favourably to those obtained by sensory analysis.

Cole and Roberts (1997) used a different approach for their measurement of translucent molten gelatine solutions. Considering that the light which reaches the human eye during visual assessment of gelatine samples comprises both reflected and transmitted light, Cole and Roberts used a spectrophotometer and a transreflectance presentation of the samples. In such cases, it is essential that the scatter within the sample (in this case caused by molecular gelatine) is small in comparison to that from the white background tile. Tristimulus Hunter *L* data was found to correlate to colour values ascribed via Beer's law and a DCU index often used in the gelatine industry. The authors suggest that such a reference instrumental method of measurement could overcome many of the problems which occur as a result of the numerous 'in-house' colour grading systems currently in use.

Francis and Clydesdale (1972) urged that egg yolks should be homogenised before colour measurement. This is because yolks are formed in a layered manner and poultry feed variations can, therefore, cause colour variations within each yolk. Desarzens *et al.* (1983) had a large capacity sample cell especially made to use with the MacBeth MS-2000 for liquid milk products. In a study of several types of liquid milk products, the authors were able to establish a relationship between measured colour changes in the products and a decrease of riboflavin content during storage exposure to light. MacDougall (1987) has shown that instrumental variables, such as aperture size and illumination area, have a large effect on the colour values obtained for a number of translucent liquids.

4.5 Conclusions

It is clear that to be able to compare colour measurements on different samples of the same food materials, either within a laboratory or between different laboratories, the experimental procedure must be carefully defined and standardised. Despite the obviousness of this statement, it is surprising how little information is given in much of the available literature on colour measurement, concerning the following essential areas:

- the choice of the viewing geometry and illumination type (which is associated with the selection of the instrument)

- the calibration of the instrument, including defining the calibration standards
- the selection of the instrument variables of aperture and illumination areas
- the selection, preparation and presentation of the samples
- the defining of other relevant details of the test procedure, such as the temperature.

Hunter (1987) has given a very useful checklist, in standardised format, which can be used as a reminder when documenting the variables in an experimental procedure for measuring the colour of a food material.

When at a loss as to how to proceed with a new problem in food colour measurement, it may be as well to keep in mind a number of pieces of good advice given by Francis and Clydesdale (1975):

1. To measure the visual impact of a food sample, the golden rule is to perform the measurement on the sample as the consumer sees it, with as little sample preparation as possible.
2. To predict the content of a chemical colour component in a food material by colorimetry, it is best to ensure when setting up the predictive curve that the only variable is the colour component.
3. Having obtained a good correlation between a particular colour function and the visual judgement in one application it is not safe to assume, without further visual testing, that the colour function will give an equally good correlation in the case of another food material.

With advancing technology and decreasing hardware costs, colour reflectance measurement is becoming a much more accessible tool for the whole of the food industry. Using instrumental colour measurement in combination with sensory analysis and other physical measurements can provide insights into many phenomena associated with food manufacture. The ever-growing choice of instruments on the market means that the best purchase choice can be made depending upon the samples to be measured and the desired results. Finally, the advent of in-line analysis facilitates the advent of colour measurement as an important tool for quality control and process monitoring in the future.

4.6 References

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